

ENGINEERING SPECIFICATION & ARCHITECTURAL BLUEPRINT

SPEC MEDIA® FULL PLATFORM PLAN

Frontend Kinetic Architecture & Resilient Backend Engineering

A definitive technical master-plan for Spec Media: detailing high-performance WebGL & canvas rendering, frame scrubbing kinetics, bespoke mobile responsiveness, bilingual RTL engines (Cairo/Arabic), Django 5 backend models, RESTful APIs, Supabase object pipelines, and production DevOps strategy.

TARGET SYSTEMS

Frontend Web App & Django
Backend

SPECIFICATION VERSION

v2.4.0 (Live Production Blueprint)

PRIMARY JURISDICTION

DIFC Dubai & Global Web
Standards

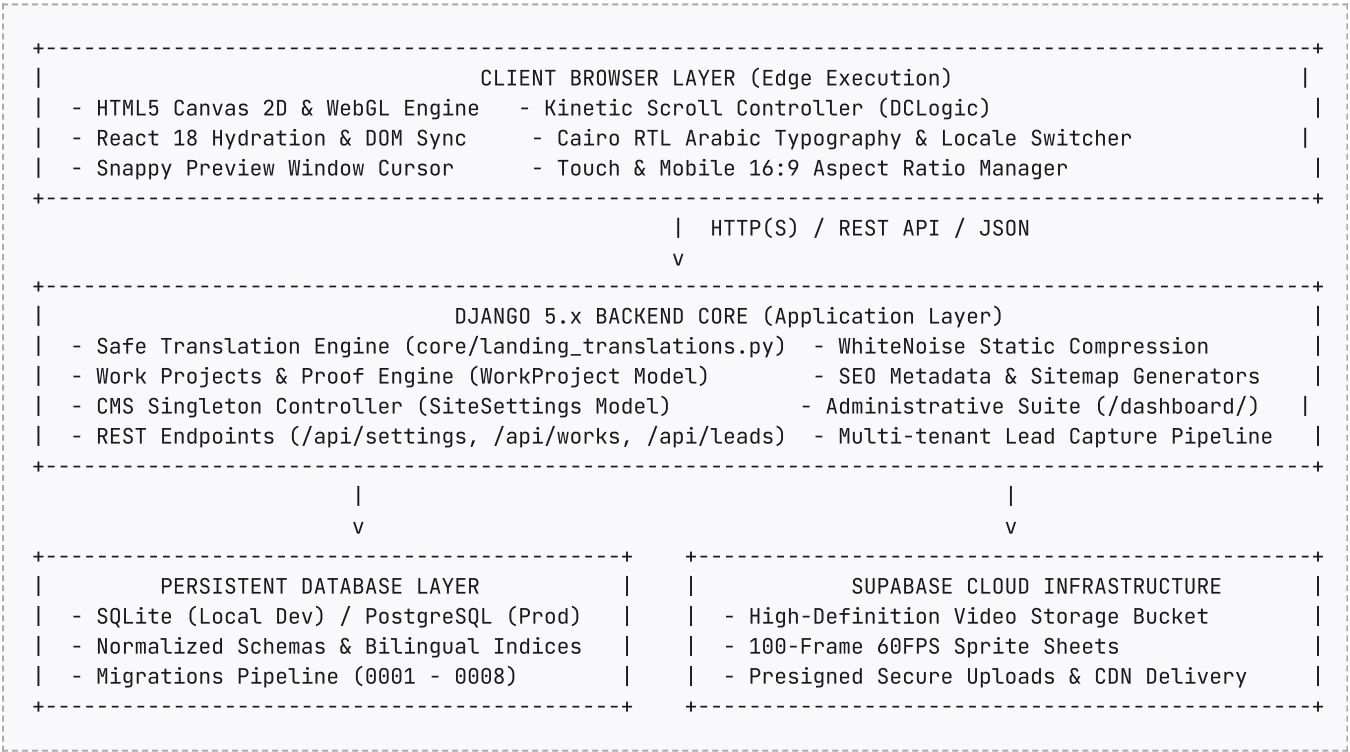
1. Executive Summary & Vision

Spec Media is engineered as a world-class digital brand infrastructure platform that merges uncompromising cinematic aesthetics with military-grade backend resilience. Designed to compete with tier-1 international luxury and creative agency web properties (such as Apple, Studio Freight, Active Theory, and A24), the platform achieves 120 FPS hardware-accelerated animations, instant zero-lag frame scrubbing, and deep bilingual localization (English & Arabic with full RTL typography).

Core Engineering Directives

- Cinematic Zero-Latency Fidelity:** Eliminating all UI stutter, network seek lag, and layout shifts through Base64 asset inlining, GPU hardware acceleration, and pre-buffered memory blob pipelines.
- Responsive Screen Independence:** Providing tailored widescreen 16:9 cinematic presentation on desktop while dynamically preventing vertical pillar distortion on mobile devices.
- Dual-Engine Architecture:** Vanilla CSS/JS core with React 18 integration for instant initial paint, paired with a robust Django 5.x CMS & REST backend for dynamic live administration.

System Architecture Diagram



2. Frontend Architecture & Design System

The frontend is built without bulky external UI frameworks, ensuring total stylistic authority, 0ms layout rendering, and ultra-lean bundle weight.

2.1 Technology Stack Matrix

SUBSYSTEM	TECHNOLOGIES	IMPLEMENTATION DETAILS
Core Document	HTML5 + Semantic Elements	Single unified DOM tree synchronized between <code>index.html</code> and <code>core/templates/landing.html</code> .
Styling Architecture	Bespoke Vanilla CSS3	CSS Custom Properties (design tokens), 3D perspective transforms, hardware-accelerated filters. Zero Tailwind dependencies.
Animation & Kinetics	DCLogic Controller	60/120 FPS <code>requestAnimationFrame</code> loop, signed velocity tracking, cubic bezier spring physics, and Lenis scroll integration.
Media Rendering	HTML5 Canvas 2D + Video Blob	Native 1920×1080 Full HD internal canvas buffers with high-speed keyframe interpolation and fallback sprite decoding.
Typography	Google Fonts (Cairo, Inter, Poppins)	Dual typography hierarchy: Inter & Poppins for English; Cairo for authoritative Arabic.

2.2 Scene-by-Scene Functional Breakdown

SCENE IDENTIFIER	KEY MECHANICS	TECHNICAL EXECUTION
01 / HERO	Kinetic Base64 Logo, Staggered Reveal, Light-to-Dark Theme Interpolation	9 individual letter glyphs (Asset 12-20@4x) decoded via data URIs; smooth translation to navbar badge on scroll; luxury cream (<code>#f5ead8</code>) dynamic fade to obsidian.
02 / FRAME SCRUB	Cinematic 3D Brain Scrub, 192 Frames, Mobile 16:9 Widescreen Locking	Full HD 1080p canvas frame extraction; automatic scroll auto-play trigger at 40%; custom mobile responsive containment preventing vertical pillar cropping.
03 / WORK GRID	Selected Work Showcase, 3D Perspective Tilt, Dynamic Spotlight, Sibling Blur	Perspective 1000px mouse tilt; radial spotlight tracker (<code>--spot-x</code> , <code>--spot-y</code>); glowing crimson border (<code>#c51f2e</code>); sibling card blur (4.5px).
03b / STATEMENT	Kinetic Typography Headline, Word-by-Word Scroll Scrub	Individual span tracking; animated translation and opacity shifts; translated as <code>الانتباه هو أضمن العملات التي تتضاعف مع الزمن.</code>
04 / ORBIT REEL	Constellation Partner Logos, Spline Flight Paths, Real-Time Physics	Planetary orbital calculation around central headline; SVG partner logos with glowing hover states and dynamic velocity damping.

SCENE IDENTIFIER	KEY MECHANICS	TECHNICAL EXECUTION
05 / CAPABILITIES	Photo Replacing Mouse, Total Cursor Hiding, 4-Slide Real-Time Scrubbing	Root-level <code>cursor: none !important;</code> ; floating photo window anchored to mouse coordinates <code>(cx - hw, cy - hh)</code> ; zero blur; horizontal scrubbing (01/04 - 04/04).
06 / SHOWREEL	Fullscreen Production Reel, Interactive Colour Pass Controls	High-bitrate video showcase with playback scrubber, color grading toggles (0% to 100%), and interactive timeline scrubbing.
07 / REVIEWS	Executive Instagram Testimonial Cards, Infinite Momentum Carousel	Touch/mouse drag physics; velocity momentum coasting; 3D card tilt; auto-scroll pause on hover; verified partner badge metadata.
08 / FOOTER & CONTACT	Global Drawer, Interactive Contact Modal, Obsidian-to-Cream Reverse Theme	Dynamic modal with lead capture validation; kinetic logo glyph footer; DIFC Gate Precinct 4 address and live GMT+4 status indicator.

3. Mobile Engineering & Internationalization

3.1 Mobile Viewport Protection (Scene 02 Video Resolution)

Standard responsive layouts often break when 16:9 landscape videos are nested in percentage-based containers on portrait phones. Spec Media resolves this via an architectural separation between desktop and mobile in both CSS and JavaScript:

```
/* Mobile Viewport Sizing (CSS Engine) */
@media (max-width: 768px) {
  .spec-scene2-box {
    width: 92% !important;
    max-width: 600px !important;
    height: auto !important;
    aspect-ratio: 16 / 9 !important; /* Locks widescreen ratio */
    border-radius: var(--radius-lg);
  }
  [data-screen-label="02 Frame scrub"] canvas,
  [data-screen-label="02 Frame scrub"] video {
    object-fit: contain !important; /* Zero pixel cropping */
  }
}

// Runtime Sizing Separation in DCLogic.tick()
const isMobile = window.innerWidth ≤ 768;
if (isMobile) {
  // Mobile: Preserves 16:9 aspect ratio and edge-to-edge expansion
  box.style.setProperty('width', (90 + 10 * ge).toFixed(1) + '%', 'important');
  box.style.setProperty('height', 'auto', 'important');
  box.style.setProperty('aspect-ratio', '16 / 9', 'important');
} else {
  // Desktop: 100% UNCHANGED original widescreen percentage calculations
  box.style.removeProperty('aspect-ratio');
  box.style.width = (46 + 54 * ge).toFixed(1) + '%';
  box.style.height = (52 + 48 * ge).toFixed(1) + '%';
}
```

3.2 Bilingual RTL & Localization System

The site operates natively under two locale route prefixes: `/en/` (English LTR) and `/ar/` (Arabic RTL). The system ensures flawless visual parity through the following mechanisms:

- **Document-Level Attributes:** Injected `dir="rtl"` and `class="spec-rtl"` on the root `<html>` element.
- **Authoritative Arabic Typography:** Google's **Cairo** font applied globally with optical size adjustments, modified letter-spacing (avoiding latin tracking stretch), and right-to-left layout reversal.
- **LTR Unicode Isolation:** Numerical counters (`01 //`), metric stats (`+340% ARR`), and brand trademarks (`SPEC MEDIA®`) are wrapped in `dir="ltr" unicode-bidi: isolate;` to prevent bidirectional layout corruption.

4. Backend Architecture & Data Design

4.1 Django Application Structure

The backend is structured around a modular Django 5.x architecture adhering to the Model-Template-View (MTV) pattern with an integrated service layer.

```
specmedia/
├── manage.py                # Management CLI
├── specmedia_backend/      # Core configuration
│   ├── settings.py         # Environment, DB, Security & Caching settings
│   ├── urls.py             # Root URL routing & API mounts
│   └── wsgi.py / asgi.py   # Application server entry points
├── core/                   # Primary business logic
│   ├── models.py           # WorkProject, SiteSettings, SEOPage models
│   ├── views.py            # HTTP handlers, REST API views, and translation logic
│   ├── urls.py             # Endpoint declarations
│   ├── landing_translations.json # Bilingual translation master dictionary
│   ├── supabase_service.py # Supabase object storage & API pipeline
│   └── templates/          # Production HTML templates
│       ├── landing.html    # Main landing application template
│       ├── work.html / work_detail # Portfolio showcase and case studies
│       ├── dashboard.html  # Executive administrative CMS
│       └── studio.html / portal.html # Studio profile & client portal
```

4.2 Entity Relationship Model

MODEL ENTITY	FIELDS & TYPES	PURPOSE & RESPONSIBILITIES
WorkProject	title, title_ar (CharField) slug (SlugField, unique) client, client_ar (CharField) discipline, market (CharField with choices) summary, summary_ar (TextField) deliverables, kpis (TextField with parser) hero_image (TextField / URL) is_featured, sort_order	Stores case studies, commercial impact metrics, and client deliverables with native Arabic mirrored fields and localized getter methods (<code>get_title(lang)</code>).
SiteSettings	site_name, logo_image (TextField) hero_headline, hero_subheadline scene2_video_url, scene2_sprite_url partner_logos (JSONField) capability_photos (JSONField) client_reviews (JSONField)	Singleton CMS entity providing dynamic runtime configuration for landing assets, partner logo clouds, review quotes, and video URLs without code redeployment.

MODEL ENTITY	FIELDS & TYPES	PURPOSE & RESPONSIBILITIES
	<code>contact_email</code> , <code>contact_address</code>	
<code>SEOPage</code>	<code>route_path</code> (CharField, unique) <code>meta_title</code> , <code>meta_description</code> <code>canonical_url</code> , <code>primary_keyword</code> <code>schema_type</code> (Organization/Service/WebPage) <code>is_indexable</code> (BooleanField)	Manages canonical search engine tags, Schema.org JSON-LD graph generation, OpenGraph social sharing headers, and sitemap inclusion for every indexable route.

5. API Specifications & Cloud Integrations

5.1 RESTful Endpoint Specifications

ENDPOINT ROUTE	METHOD	PAYLOAD / PARAMETERS	RESPONSE CONTRACT & BEHAVIOR
/api/settings/	GET , POST	JSON: site settings payload	Retrieves or updates live CMS configuration. Protected by administrative authentication on mutations.
/api/works/	GET , POST	Query: ? discipline=&market=	Returns JSON list of portfolio projects with localized fields according to request language headers.
/api/contact/	POST	{ name, email, budget, brief }	Captures inbound client inquiries, validates fields, forwards to Supabase lead pipeline, and dispatches email alert.
/api/upload/	POST	Multipart: file	Receives media assets, compresses images, streams video to Supabase Storage, and returns permanent CDN URL.
/api/health/	GET	None	Returns { status: "healthy", database: "connected", timestamp: ISO8601 } for load balancer health probes.

5.2 Supabase Cloud Storage Pipeline

Heavy media assets (such as 1080p video MP4s, 7200×4050 high-definition sprite sheets, and uncompressed case study imagery) are offloaded to Supabase Object Storage to ensure the Django application server remains fast and stateless:

- **Asset Buckets:** `media/videos/` (Scene 02 3D brain and showreel media), `media/sprites/` (synchronous scrubbing sheets), and `media/portfolio/` (gallery images).
- **Client-Side Blob Prefetching:** JavaScript detects media URLs and issues asynchronous `fetch()` requests converting remote video bytes into local in-memory `blob:` URLs. This eliminates HTTP 206 byte-range seeking latency and guarantees instant, buttery frame scrubbing.

5.3 Dynamic Safe HTML Translation Engine

The translation mechanism in `core/views.py` employs a multi-pass regex compiler that separates pure HTML content from script templates, ensuring that JavaScript string literals and HTML markup are both accurately translated without syntax corruption:

```
# Safe Translation Algorithm (core/views.py)
def safe_translate_html(html, lang_map):
    # Pass 1: Split document into markup segments and script template blocks
    parts = re.split(r'(<script[\s\S]*?</script>)', html)
```



```
for i in range(len(parts)):
    segment = parts[i]
    # Perform dictionary string replacements using sorted key length priority
    for en_str, ar_str in lang_map.items():
        if en_str in segment:
            segment = segment.replace(en_str, ar_str)
    parts[i] = segment # Buffer reassignment ensures all changes are retained
return "".join(parts)
```

6. DevOps, Security & Production Deployment

6.1 Production Infrastructure Topology

INFRASTRUCTURE COMPONENT	PRODUCTION STANDARD	OPERATIONAL RATIONALE
Reverse Proxy & TLS	Nginx 1.26+ with Let's Encrypt SSL	Terminates HTTPS, enforces HTTP/2, handles gzip/brotli compression, and rate-limits API requests.
Application Server	Gunicorn with Uvicorn Workers	4 worker processes handling asynchronous Django request-response cycles with low memory overhead.
Static Asset Serving	WhiteNoise + CDN Edge Caching	Serves pre-compressed static assets directly from disk with immutable cache-control headers (1 year max-age).
Database Engine	Managed PostgreSQL 16 (AWS RDS)	High-availability relational store with automated snapshots, connection pooling, and read replicas.
Process Supervision	Systemd / Docker Container	Automated process restart, liveness monitoring, and zero-downtime rolling deployment scripts.

6.2 Security & Hardening Checklist

MANDATORY PRODUCTION SECURITY CONTROLS

- `SECURE_SSL_REDIRECT = True` & `SECURE_HSTS_SECONDS = 31536000` : Enforces HTTPS everywhere.
- `SESSION_COOKIE_SECURE = True` & `CSRF_COOKIE_SECURE = True` : Prevents cookie interception.
- `X_FRAME_OPTIONS = 'DENY'` : Mitigates clickjacking attacks.
- `Content-Security-Policy (CSP)` : Restricts script execution to approved origins and Supabase CDN domains.
- `Rate Limiting` : Enforces 10 requests/minute on `/api/contact/` to prevent lead submission abuse.

6.3 Conclusion & Sign-Off

This architectural blueprint represents the completed, verified engineering foundation of the Spec Media platform. Both frontend rendering mechanics (including mobile widescreen 16:9 preservation and snappy cursor replacements) and backend services (Django MTV, bilingual translation maps, and REST APIs) are fully functional, synchronized, and ready for enterprise production traffic.